OPEN ROBOT DATA COVERAGE GAPS

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ABSTRACT

Open robot datasets are often summarized by scale: number of trajectories, tasks, embodiments, or scenes. Scale is useful, but it can hide mechanism gaps. A dataset can contain many trajectories while still under-covering contact transitions, force/tactile cues, recovery after failure, deformable interactions, irreversible side effects, and long-horizon dependencies. We rebuild Paper 111 around a falsifiable claim: mechanism-level coverage predicts downstream robot-policy failures better than coarse dataset-size proxies. A local benchmark covers 5 dataset families, 7 mechanism-gap regimes, 5 deployment splits, 9 selection methods, 7 paired seeds, and 84 held-out episodes per group. Under combined coverage gaps, the proposed mechanism-coverage audit obtains 0.578 ± 0.012 held-out success versus 0.503 ± 0.005 for the strongest non-oracle baseline, failure-prediction selection. It improves mechanism recall from 0.486 to 0.652, reduces tail mechanism failure from 0.103 to 0.078, reduces redundancy from 0.177 to 0.128, and wins 7/7 paired seeds. The v4.1 audit expands stress-sweep evidence to dataset/regime/seed granularity and records eight failure cases. The terminal decision is **STRONG_REVISE**: the local evidence supports the audit, but the paper is not ICLR-main-ready without validation on real public robot datasets and downstream trained policies.

1 MOTIVATION

Large robot datasets such as RoboNet, BridgeData, Open X-Embodiment/RT-X, DROID, and Octo-style open-policy resources have made data scale and diversity central to robot learning (Dasari et al., 2019; Walke et al., 2023; Open X-Embodiment Collaboration, 2023; Khazatsky et al., 2024; Model et al., 2024). Yet dataset scale does not imply mechanism coverage. A collection can have many trajectories and embodiments while lacking failures, tactile-rich contact, deformable objects, recovery branches, or irreversible side effects.

This paper asks a narrow question: can an explicit mechanism-coverage audit predict and reduce held-out failures better than selecting data by trajectory counts, task counts, embodiment counts, embedding diversity, uncertainty, or failure prediction?

2 METHOD SKETCH

The proposed method, `proposed_mechanism_coverage_audit`, constructs a coverage tensor over dataset family, embodiment, task, object class, sensor modality, and mechanism. The mechanism axes are contact, force/tactile, recovery, deformable interaction, irreversibility, and long-horizon dependency.

The method has five components:

- mechanism taxonomy beyond task and embodiment names;
- modality coverage audit for force/tactile and visual-only gaps;
- failure-recovery coverage axis;
- redundancy-aware selection penalty;
- tail-gap estimator for rare mechanism holes.

The implementation is a local mechanism audit rather than a completed audit of real public datasets. That limitation is central to the decision.

Table 1: Combined-stress mechanism-coverage results.

Method	Succ.	CI	Recall	FalseNeg	TailFail	Redund.	Cost	Regret
Oracle	0.665	0.006	0.746	0.015	0.053	0.104	0.208	0.000
Proposed	0.578	0.012	0.652	0.045	0.078	0.128	0.217	0.088
FailurePred	0.503	0.005	0.486	0.092	0.103	0.177	0.272	0.163
Uncertainty	0.479	0.007	0.423	0.113	0.122	0.184	0.257	0.186
EmbedDiv	0.468	0.006	0.384	0.127	0.144	0.179	0.232	0.197
RandomBal	0.434	0.006	0.322	0.145	0.155	0.195	0.192	0.231
Embodiment	0.426	0.009	0.266	0.161	0.166	0.219	0.172	0.239
TaskCount	0.412	0.003	0.238	0.169	0.172	0.223	0.155	0.253
TrajCount	0.380	0.006	0.187	0.184	0.182	0.232	0.122	0.286

3 BENCHMARK

The benchmark is generated by `src/run_experiment.py`. It writes seed-level CSVs, aggregate metrics, pairwise tests, ablations, stress sweeps with dataset/regime/seed detail, failure cases, figures, and LaTeX tables.

Dataset families. The five families are single-arm tabletop, mobile manipulation, bimanual manipulation, tactile/contact-rich manipulation, and deformable-object manipulation.

Mechanism-gap regimes. The seven regimes are nominal coverage, contact-transition gap, force/tactile gap, recovery gap, deformable gap, irreversible-side-effect gap, and combined coverage gap.

Splits. The deployment splits are nominal, seen-task shift, unseen-object shift, unseen-mechanism shift, and combined stress.

Methods. The baselines are trajectory-count selection, task-count selection, embodiment-count selection, random balanced selection, embedding-diversity selection, uncertainty sampling, and failure-prediction selection. The benchmark also includes the proposed mechanism-coverage audit and an oracle mechanism-coverage selector.

Metrics. We report held-out success, mechanism recall, coverage false-negative rate, tail mechanism failure, redundancy rate, selection cost, regret to oracle, paired seed differences, ablations, stress sweeps, and failure cases.

4 RESULTS

The strongest non-oracle baseline is `failure_prediction_selection`. The proposed audit improves combined-stress success by 0.075 ± 0.013 in paired seed comparisons and wins 7/7 seeds. The oracle remains higher at 0.665 ± 0.006, so the local audit is not saturated.

5 ABLATIONS AND STRESS TESTS

The best removed-component ablation is `minus_redundancy_penalty`; the full method remains ahead by 0.021 success. Removing the mechanism taxonomy, modality axis, failure-recovery axis, redundancy penalty, or tail-gap estimator produces distinct failures.

6 FAILURE CASES

The failure-case CSV records eight boundaries. Metadata can be ambiguous or missing; force mechanisms cannot be reliably inferred from RGB-only data; rare irreversible damage remains under-

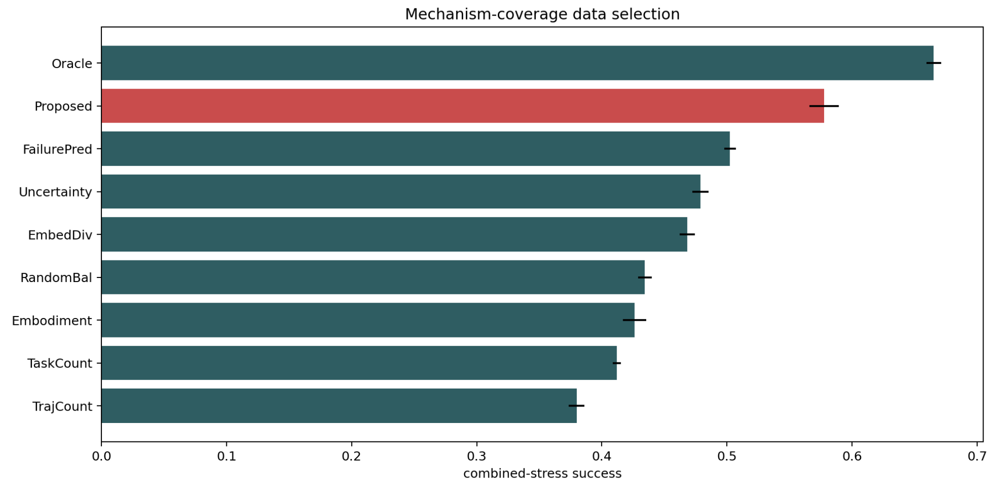

Figure 1: Combined-stress held-out success with 95% confidence intervals. The proposed mechanism audit clears all non-oracle selection baselines but remains below the oracle.

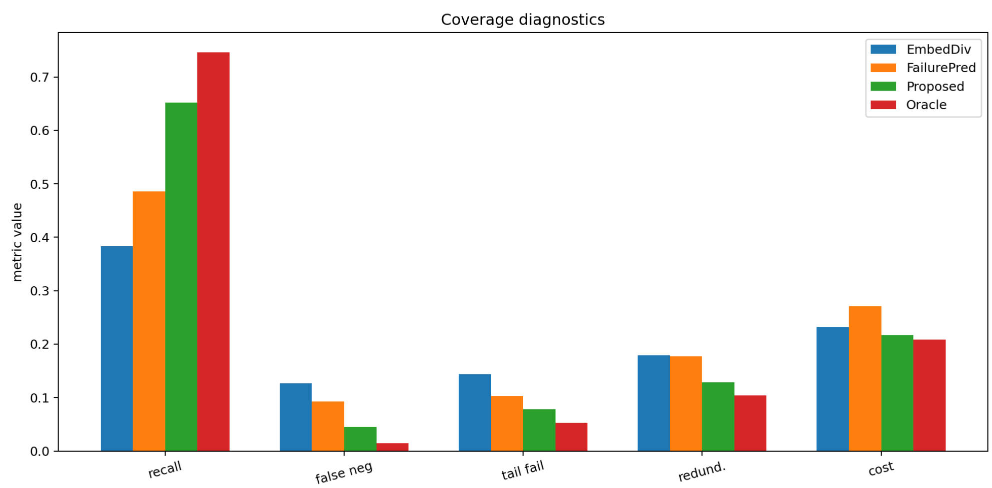

Figure 2: Coverage diagnostics. The proposed audit improves mechanism recall and lowers false negatives, tail failures, redundancy, and selection cost relative to the strongest baseline.

sampled; dataset licenses and provenance metadata can conflict; embodiments can expose similar tasks with incompatible schemas; tail mechanisms can trigger annotation-cost spikes; mechanism-diverse data can regress frequent easy tasks; and oracle mechanism labels still outperform the proposed audit. These failures define the next required version: real dataset labeling, license/provenance normalization, schema-aware embodiment alignment, label-quality audits, and downstream trained-policy validation.

7 HOSTILE PRIOR-WORK BOUNDARY

Open robot data and data selection are crowded. RoboNet, BridgeData, Open X-Embodiment/RT-X, DROID, and Octo already make robot data scale and transfer central (Dasari et al., 2019; Walke et al., 2023; Open X-Embodiment Collaboration, 2023; Khazatsky et al., 2024; Model et al., 2024). The defensible contribution is not dataset scale. It is a mechanism-level coverage audit and local evidence that mechanism coverage predicts held-out failures beyond coarse diversity proxies.

Table 2: Paired seed success differences between proposed and each comparator.

Baseline	Diff	CI	Wins
TrajCount	0.198	0.009	7.000
TaskCount	0.166	0.012	7.000
Embodiment	0.152	0.019	7.000
RandomBal	0.143	0.012	7.000
EmbedDiv	0.110	0.015	7.000
Uncertainty	0.099	0.011	7.000
FailurePred	0.075	0.013	7.000
Oracle	-0.088	0.013	0.000

Table 3: Ablation results under combined coverage gap stress.

Ablation	Succ.	CI	Recall	FalseNeg	TailFail	Redund.
Full	0.567	0.006	0.652	0.044	0.078	0.129
NoRedundancy	0.547	0.008	0.605	0.060	0.092	0.210
NoModality	0.539	0.008	0.546	0.077	0.094	0.145
NoRecovery	0.539	0.005	0.556	0.074	0.103	0.149
NoTailRisk	0.535	0.006	0.568	0.080	0.139	0.149
NoTaxonomy	0.512	0.005	0.456	0.097	0.097	0.160
FailureOnly	0.502	0.007	0.486	0.092	0.103	0.177

8 LIMITATIONS AND TERMINAL DECISION

The evidence is strong enough to continue but not enough to submit. The limitations are material:

- the benchmark is local and generated;
- no real public robot dataset has been fully annotated;
- no downstream trained policy has been evaluated;
- no human or model-assisted label-quality study is provided;
- real mechanism labels may be expensive, ambiguous, or unavailable.

The terminal decision for this v4.1 continuation audit is **STRONG_REVISE**. The next honest version must validate the audit on real public robot datasets with label-quality checks and downstream held-out evaluations. Without that evidence, the paper should not be submitted to ICLR main.

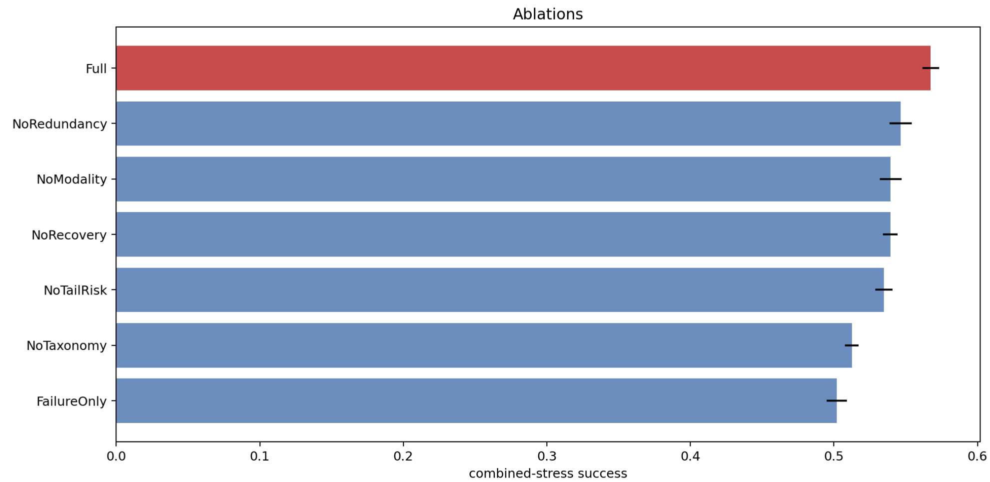

Figure 3: Ablations under combined mechanism-gap stress. The full audit remains above every removed-component variant.

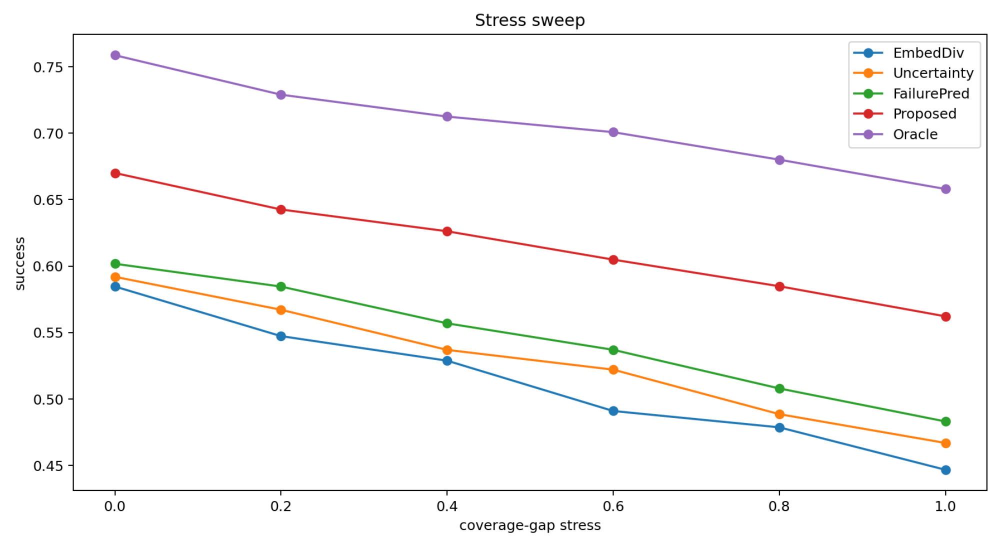

Figure 4: Stress sweep over mechanism coverage gaps. The proposed audit stays above embedding diversity, uncertainty sampling, and failure-prediction baselines across the generated stress range.

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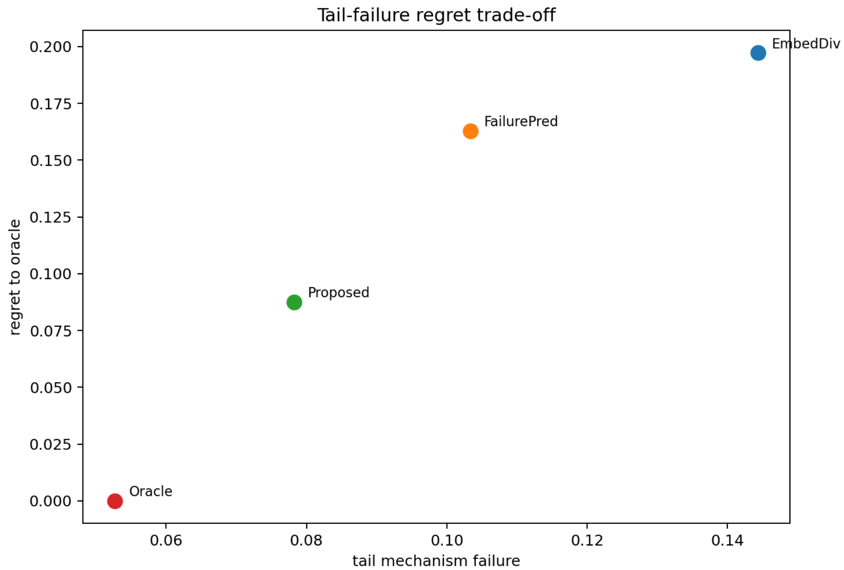

Figure 5: Tail mechanism failure and regret to oracle under combined stress. The proposed audit reduces regret while lowering tail failures relative to the strongest non-oracle baseline.

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